

CPU Utilization as a Software-Only Thermal Proxy for Multi-Tenant Edge AI Inference: A 13-Hour Characterization and Coupling Law on Raspberry Pi 5

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Abstract—Thermal-aware control of edge AI inference usually presumes instrumented power sensing, which most deployed single-board computers (SBCs) lack. We ask a narrower, practical question: on a current-generation SBC running real inference, how well does *CPU utilization alone*, a fully software-observable signal, predict SoC temperature, and is the relationship stable enough to control against? From a 13-hour, four-block campaign on Raspberry Pi 5 (232,136 samples at 5 Hz: characterization, load sweeps, eight two-tenant configurations, and a six-hour stability run) we obtain a cross-validated *CPU-thermal coupling law*, $T \approx 45.6^\circ\text{C} + 0.175^\circ\text{C}/\% \cdot U$ ($R^2 = 0.93$), with leave-one-block-out error of $0.84\text{--}2.45^\circ\text{C}$ across four blocks of differing workload composition, a first-order die response of $\tau \approx 1\text{--}2\text{ s}$, and a slope that transfers across a six-month recalibration within 5%. Compared like-for-like against the standard structures (lumped-RC, autoregressive, and multi-signal models), the law matches every sensor-free, actuatable alternative (all collapse to $1.6\text{--}1.8^\circ\text{C}$), while sensor-feedback forecasters are $2.7\times$ tighter at one step but cannot answer allocation what-ifs and presuppose the sensor. The law converts a temperature budget into a utilization budget with no power instrumentation. We release the runtime and full dataset.

Index Terms—Edge computing, edge AI, thermal management, Raspberry Pi, empirical characterization, reproducible research, ONNX Runtime.

I. INTRODUCTION

EDGE AI deployments increasingly run continuous neural-network inference on small single-board computers (SBCs) in always-on perception pipelines. These platforms operate under a junction-temperature ceiling enforced by modest cooling, a fixed pool of cores shared among concurrent workloads, and application latency requirements. The thermal constraint is distinctive because it integrates load over time: sustained inference heats the SoC toward a firmware throttle threshold beyond which frequency is cut and latency degrades nonlinearly.

Thermal-aware runtime control requires a model that predicts temperature from observable signals. The natural predictor is board power, and a large body of work assumes an

instrumented power rail; most deployed SBCs ship none and cannot be retrofitted in the field. The practical question for a deployable controller is thus “*what can I predict temperature from using only software-observable telemetry?*”

This paper answers that question empirically for Raspberry Pi 5, the current-generation reference SBC. We show that CPU utilization, available from the operating system with no special hardware, predicts SoC temperature with $R^2 = 0.93$ through a simple linear law that is stable across workloads and time, and whose dynamics are fast enough for real-time use. Concretely, we make four contributions:

- **A cross-validated coupling law** for Raspberry Pi 5, $T \approx 45.6 + 0.175 U$, validated leave-one-block-out across four blocks of differing model, resolution, request rate, and tenancy (§IV).
- **Its dynamics**: a die response of $\tau \approx 1\text{--}2\text{ s}$ under a minutes-scale heatsink component, fast enough that the steady-state law is the right model at second-scale control ticks (§V).
- **A like-for-like comparison** against lumped-RC, AR/ARX, and multi-signal models under one protocol (§VI).
- **A multi-tenant characterization** over eight two-tenant pairs, and an open runtime plus 232,136-sample dataset (§VII, §IX).

We are explicit about scope. We report no board-power measurements: the measurement platform had no power sensor attached, and we present the coupling law as the software-only alternative that this absence motivates. We also note that the CPU frequency governor was pinned to performance (constant 2.4 GHz), so the law characterizes the fixed-frequency regime; we return to both points in §X.

II. RELATED WORK

DNN inference on edge SBCs. Efficient architectures [1], optimized runtimes [2], and edge benchmarks [5] characterize inference latency on SBCs, but none provide a reusable utilization-to-temperature model for control.

Thermal- and power-aware computing. Dynamic thermal management [3], [4] and DVFS operate below the application layer and assume privileged frequency control or instrumented

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Runtime, configurations, raw telemetry, and analysis scripts are released at <https://github.com/manunicholasjacob/pi5-thermal-proxy>.

power, neither available to an application-level controller on a stock SBC.

Software thermal and power estimation. The established structures are: performance-counter power models feeding a lumped RC thermal model, as in Isci and Martonosi’s runtime power monitoring [6] and Lee and Skadron’s counter-driven temperature sensing through HotSpot [7]; autoregressive forecasters fit to the temperature sensor itself, as in Coskun et al.’s ARMA-based proactive management [8]; and calibrated utilization-based power models on battery devices such as PowerTutor [9]. These assume either a calibrated per-platform counter-to-power model, or continuous access to the very sensor a proxy is meant to replace. What is missing for deployed SBCs is a quantitative, cross-validated utilization-to-temperature law measured on current hardware under real inference, published with its dataset. Section VI evaluates representatives of each structure above on our data under one protocol.

III. MEASUREMENT METHODOLOGY

A. Platform

Raspberry Pi 5 (Broadcom BCM2712, quad-core Arm Cortex-A76 @ 2.4 GHz), 64-bit Raspberry Pi OS, ONNX Runtime CPU execution provider. SoC temperature is read from `vcgencmd/sysfs`; CPU and memory utilization from the OS. The CPU governor is pinned to `performance`, so core frequency is a constant 2.4 GHz throughout (we verified a single frequency value across all samples); the campaign therefore characterizes the fixed-frequency operating regime typical of a latency-oriented inference deployment.

B. Runtime and telemetry

Inference is served by a modular Python runtime that hosts ONNX Runtime workers and a telemetry thread sampling temperature, per-core CPU utilization, and memory at 5 Hz, writing time-aligned rows to Apache Parquet. Each experiment block runs sequentially; a fresh session logs each block.

C. Experiment design

The campaign comprises four blocks over 13 continuous hours: **Block 1**, single-workload characterization (MobileNetV2 and ResNet-18 at two resolutions, fixed rate); **Block 2**, a MobileNetV2 load sweep populating the utilization–temperature space; **Block 3**, eight ordered pairs of two concurrent ONNX workloads (SqueezeNet, MobileNetV2, ResNet-18, EfficientNet-lite0, YOLOv5n) on four cores; **Block 5**, a six-hour continuous MobileNetV2 run testing for drift (block labels follow the released dataset’s log naming).

D. Dataset provenance and de-duplication

The raw capture contains nine Parquet files. Each experiment block was logged by two concurrent collector processes started ~ 90 s apart; both recorded the same physical run at the same 4.95 Hz effective rate. We therefore de-duplicate to one collector per run, yielding **232,136 unique time-aligned**

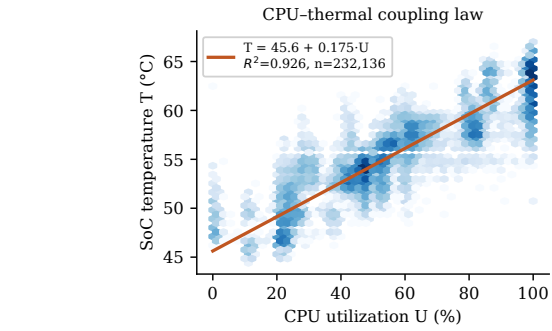


Fig. 1. CPU–thermal coupling on Raspberry Pi 5 (232,136 samples, log-density hexbins). SoC temperature is linear in CPU utilization with $R^2 = 0.93$; the fitted law is Eq. (1).

samples (the naive sum over all nine files, 471,827, double-counts). All results below use the de-duplicated set. This provenance detail is stated so that the released dataset is interpreted correctly.

E. What we measure and what we do not

We report SoC temperature, CPU utilization, and memory. We *do not* report board power: no sensor was read during the campaign, and rather than publish an unvalidated surrogate we omit power entirely (§X). Per-inference latency was not logged, so multi-tenant results (§VII) are thermal, utilization, and memory characterizations.

IV. THE CPU–THERMAL COUPLING LAW

A. Global fit

Across all 232,136 samples, SoC temperature is a strong linear function of CPU utilization (Fig. 1):

$$T = 45.63^\circ\text{C} + 0.1748 \frac{^\circ\text{C}}{\%} \cdot U, \quad (1)$$

with Pearson $r = 0.962$, $R^2 = 0.926$, and a residual standard deviation of 1.33°C . The slope is estimated extremely tightly: its 95% confidence interval is $[0.1746, 0.1750]^\circ\text{C}/\%$, a consequence of the large sample size and the genuine linearity of the relationship. Equation (1) says that each additional percent of aggregate CPU utilization raises steady-state SoC temperature by about 0.175°C , from an idle intercept near 45.6°C .

B. Cross-block validation

A law useful for control must generalize beyond the workload that produced it. We refit Eq. (1) in a leave-one-block-out protocol: train on three blocks, predict the held-out block. Prediction RMSE ranges from 0.84°C (stability block) to 2.45°C (single-workload block): one global law predicts every held-out block within 2.5°C , across blocks that differ in model, resolution, request rate, and tenancy, which is the property a fixed control model needs. Within-block slope estimates scatter more widely (0.13 – $0.21^\circ\text{C}/\%$, Fig. 2): each block occupies its own limited band of utilization (the load sweep is bimodal near idle and saturation; the multi-tenant block rarely leaves 42–100%), so a within-block fit has little

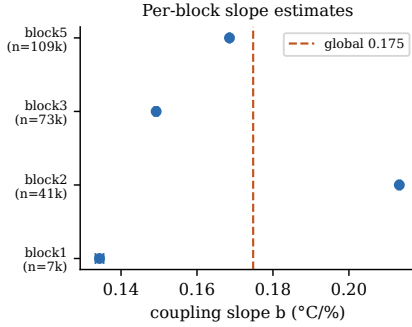


Fig. 2. Per-block slope estimates with 95% confidence intervals against the pooled fit (dashed). Within-block estimates scatter because each block spans a limited utilization band and folds ambient drift into the slope; the pooled slope is validated out of sample by held-out transfer ($\leq 2.45^\circ\text{C}$) and a six-month recalibration (within 5%).

leverage on the slope and folds slow ambient drift into it. We therefore take the pooled fit as the platform constant and validate it out of sample twice: by the leave-one-block-out transfer above, and by the independent six-month recalibration below, which reproduces the slope within 5%.

C. Portability across time and ambient

A six-month out-of-sample check: recalibrating the law on the same unit in September 2026 (different season, ambient, and background state; a 150 s three-tenant soak against a 45 s idle baseline) yields a slope of $0.183^\circ\text{C}/\%$ against the campaign’s 0.175, within 5%, while the intercept moved by only 0.1°C . The slope, which is what a controller inverts, transfers; the intercept is ambient- and cooling-dependent and should be re-measured at deployment, a 45-second idle read.

D. Interpretation for control

Equation (1) is directly actionable. A controller that knows a thermal budget $T_{\max}$ can invert the law to a *utilization budget*, $U_{\max} = (T_{\max} - 45.63)/0.1748$, with U the system-wide utilization on the operating system’s 0–100 scale. Full load projects to a steady state of 63.1°C , so a cap at the 75°C firmware-throttle level cannot bind at this ambient with stock cooling; the campaign’s observed maximum was 67.0°C (§VIII). Operator guard-band caps below 63°C do bind, and the inversion turns each into a concrete budget: $U_{\max} \approx 54\%$ for a 55°C cap. Computing it requires no power sensor: utilization is read from the OS.

V. THERMAL DYNAMICS

Equation (1) is a steady-state relationship; a controller also needs the response speed. Two estimators put the die response on the order of a second: regressing the per-sample temperature increment on the gap to the law’s steady-state target over the six-hour stability run gives $\tau \approx 0.6\text{ s}$ under a first-order model (biased low by sensor quantization at 5 Hz), and the lumped-RC grid search of §VI selects $\tau = 1\text{--}2\text{ s}$ on held-out blocks; transients decay within a few seconds. Beneath this sits a minutes-scale component: a $\sim 10\text{ s}$ exponentially-weighted

TABLE I
LEAVE-ONE-BLOCK-OUT RMSE ($^\circ\text{C}$, MEAN OVER HELD-OUT BLOCKS).
“WHAT-IF” = CAN PREDICT THE EFFECT OF A PROPOSED ALLOCATION CHANGE.

Model	Signals	What-if	1-step	Sensor-free
Coupling law (1)	U	yes	1.76	1.76
+ EWMA ($\sim 10\text{ s}$)	U	yes	–	1.64
Multi-signal static	U , mem	yes	–	1.65
Lumped RC ($\tau=1\text{--}2\text{ s}$)	U (+ T)	yes	0.67	1.71
ARX(1)	T , U	rollout	0.66	1.71
AR(1)	T	no	0.65	–

utilization improves leave-one-block-out RMSE by 0.12°C (§VI), the signature of heat-spreader mass, and sustained load moves the operating point over minutes, which is why calibration requires a soak rather than an idle read. The consequence for control: at ticks of a second or slower, an allocation’s thermal effect is substantially realized within a few ticks ($3\tau \approx 3\text{--}6\text{ s}$), so the steady-state law is the right model; only sub-second control would need the transient term.

VI. COMPARISON WITH STANDARD PREDICTION STRUCTURES

The natural question for any proxy is how it compares with the established alternatives. We implement the standard structures from §II on the released dataset and evaluate all of them under the identical leave-one-block-out protocol: a HotSpot-class first-order lumped RC model driven by utilization [7] (time constant by grid search, steady-state gain by least squares); a Coskun-class autoregressive forecaster AR(1) on the temperature sensor [8] and its ARX extension with utilization as the exogenous input; a multi-signal static regression on utilization plus memory occupancy (the counter-model structure of [6] restricted to signals a stock SBC exports); and an exponentially-filtered variant of our own law. Table I reports both tasks that matter: 200 ms-ahead forecasting *with* the temperature sensor in the loop, and sensor-free prediction from software signals alone (the input a sensorless allocation controller actually consumes), obtained from the dynamic models by rolling them out on utilization without temperature feedback.

With the sensor available, short-horizon forecasters are $2.7\times$ tighter (0.65 vs. 1.76°C at one step), but they answer the wrong question for allocation: an AR-family model has no term through which a proposed duty-cycle change enters, and it requires the sensor whose absence motivates a proxy. Every sensor-free, actuatable structure instead lands in the same $1.6\text{--}1.8^\circ\text{C}$ band. Rolling out the dynamic RC or ARX models on utilization without temperature feedback recovers no accuracy over the static law, because with $\tau \approx 1\text{--}2\text{ s}$ a rollout converges to the law’s steady state within a few seconds, and transients occupy a negligible share of the 13-hour campaign; the richer static models buy 0.1°C . On this class of platform the single-signal law is the accuracy frontier for sensor-free thermal prediction, at two constants, and the frontier is adequate for the job: a $1.6\text{--}1.8^\circ\text{C}$ error band is small against the 8°C minimum

TABLE II
MULTI-TENANT SUMMARY (BLOCK 3, EIGHT TWO-TENANT PAIRS, 73,508 SAMPLES). COUPLING-LAW PREDICTION USES EQ. (1) AT THE OBSERVED MEAN UTILIZATION.

Quantity	Value
Concurrent workload pairs	8
Mean CPU utilization	78.6 %
Mean SoC temperature (observed)	59.3 °C
Mean SoC temperature (Eq. 1 prediction)	59.4 °C
Mean resident memory	1,162 MB
Thermal-budget violations (>75 °C)	0

margin observed in §VIII and the multi-degree guard-bands operators set.

VII. MULTI-TENANT CHARACTERIZATION

Block 3 runs eight ordered pairs of two concurrent ONNX workloads on the four cores, spanning SqueezeNet, MobileNetV2, ResNet-18, EfficientNet-lite0, and YOLOv5n (73,508 samples; Table II). The coupling law predicts the multi-tenant mean temperature within 0.1 °C from the observed mean utilization ($45.63 + 0.1748 \times 78.6 = 59.4$ vs. 59.3 °C observed), evidence that the single law also governs the multi-tenant regime. No pair exceeded the thermal budget; the binding constraint under two tenants is memory and core count, not temperature.

VIII. CONSTRAINT HEADROOM

Across the 13-hour campaign, SoC temperature ranged 44.4–67.0 °C with **zero** excursions above a 75 °C budget, and the six-hour stability block showed no monotonic drift. For the workloads tested with stock cooling, thermal control on Raspberry Pi 5 is therefore *preventive* rather than reactive; and because Eq. (1) predicts the utilization at which a budget binds, that margin is maintainable from software-observable utilization alone.

IX. REPRODUCIBILITY ARTIFACT

We release the runtime, block configurations, the full de-duplicated 232,136-sample dataset, and scripts that regenerate every number, figure, and the comparison in Table I. The schema and de-duplication procedure (§III) are documented so the data is interpreted correctly.

X. LIMITATIONS

No power measurement. The campaign includes no board-power data. We note for precision that Raspberry Pi 5 does expose PMIC rail telemetry in software (`vcgencmd pmic_read_adc`); this campaign predates our use of that interface, and validating the law against PMIC power is future work. The released dataset originally carried placeholder power fields written by the collector; they contained no measurement and have been removed from the artifact, with a note documenting the removal.

Fixed frequency. The governor was pinned to performance, so Eq. (1) characterizes the constant-2.4-GHz regime. Under a dynamic governor, frequency

becomes a second thermal driver and the single-variable law would need a frequency term.

Single board and ambient. Results are for one Raspberry Pi 5 unit under laboratory ambient with stock cooling. The intercept (idle temperature) is ambient-dependent; the slope is expected to be more portable but is not verified across units or ambients here.

No per-inference latency. The telemetry does not include per-request latency, so multi-tenant interference is characterized thermally and in memory, not in latency or throughput.

XI. CONCLUSION

On Raspberry Pi 5, CPU utilization predicts SoC temperature through a tight, cross-validated linear law, $T \approx 45.6 + 0.175 U$ ($R^2 = 0.93$), predicting held-out experiment blocks within 2.5 °C, transferring across six months within 5% in slope, with a 1–2 s die response that keeps the steady-state law valid at control timescales. Compared like-for-like against the standard prediction structures, the law matches the sensor-free accuracy of lumped-RC and ARX models rolled out on utilization (the dynamics add nothing at this time constant), while sensor-feedback forecasters, though $2.7\times$ tighter at one step, cannot answer allocation what-ifs and presuppose the sensor. The law lets an application-level controller invert a temperature budget to a utilization budget from software-observable telemetry alone, with no power sensor of the kind most deployed SBCs lack. We release the runtime and the full 232,136-sample dataset to support reproducible, sensor-free thermal-control research on commodity edge hardware.

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